

ECO 362: Financial Investments

Life-Cycle Theory

Motohiro Yogo

Princeton University

Human capital

- Annual labor income has idiosyncratic risk with expected value $\mathbb{E}[L]$.
- You plan to work T more years.
- What is the present value of future labor income (a.k.a. human capital)?

$$H = \sum_{t=1}^T \frac{\mathbb{E}[L]}{(1 + Y_b(t))^t} = \mathbb{E}[L] P_a(T)$$

where

- $Y_b(t)$ is t -year zero-coupon yield.
- $P_a(T)$ is T -year annuity with \$1 payoff in each year.

Example 1: Human capital by age

- Expected annual income of \$55,000.
- Retirement at age 65.
- Use the zero-coupon yield curve on 12/31/2021.

Age	Years to retirement	Human capital (\$)
35	30	1,258,437
45	20	927,800
55	10	510,522
65	0	0

- Human capital is like a large non-tradable bond.

Idiosyncratic risk and insurance

- Unemployment.
 - Unemployment insurance funded by payroll taxes while working.
- Long-term disability.
 - Social Security disability insurance.
 - Group disability insurance through your employer.
 - Individual disability insurance from an insurance company.
- Death.
 - Important risk if you have dependents.
 - Group life insurance through your employer.
 - Individual life insurance from an insurance company.

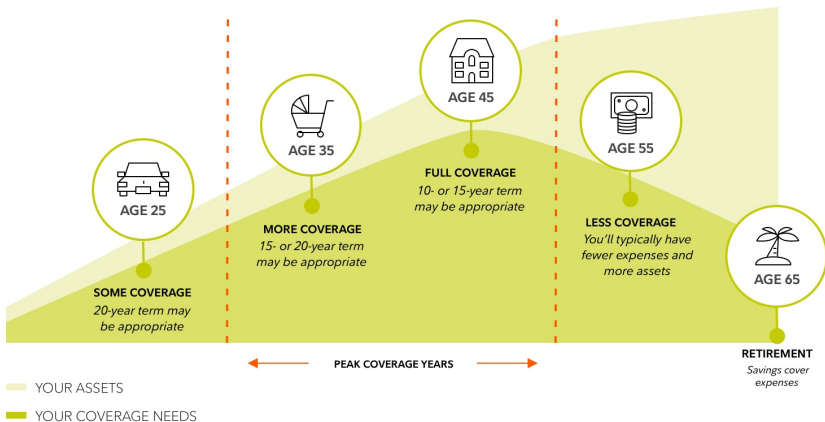
2008 mortality table

Age	Probability of death (%)	
	Male	Female
20	0.070	0.028
30	0.026	0.018
40	0.035	0.027
50	0.063	0.042
60	0.184	0.086
70	0.534	0.280
80	1.703	0.762

- At age 20, 7 out of 10,000 males and 2.8 out of 10,000 females live to age 21.

How much term life insurance?

Your expenses change throughout your life—
so will your coverage needs.



For illustrative purposes only

Source: Fidelity Investments

Life insurance

- Annual renewable life insurance.
 - Pay premium P today.
 - Insurance company pays face amount F if policyholder dies within the next year.
- Probability of death is q .
- Insurance is actuarially fair if

$$P = \frac{qF}{1 + R_b}$$

where R_b is the riskless rate.

What is the optimal face amount?

- Investor has financial wealth A , which earns riskless interest R_b .
- Can buy P dollars of actuarially fair life insurance, which pays F conditional on death.
- Conditional on being alive, H is the remaining human capital.
- Investor chooses F to maximize mean-variance expected utility over future wealth.

$$\max_F \mathbb{E}[W] - \frac{\gamma}{2} \text{Var}(W)$$

Expected wealth

- Future wealth if alive (with probability $1 - q$).

$$\begin{aligned}W(\text{alive}) &= (A - P)(1 + R_b) + H \\ &= A(1 + R_b) - qF + H\end{aligned}$$

- Future wealth if dead (with probability q).

$$\begin{aligned}W(\text{dead}) &= (A - P)(1 + R_b) + F \\ &= A(1 + R_b) - qF + F\end{aligned}$$

- Expected wealth is

$$\begin{aligned}\mathbb{E}[W] &= (1 - q)W(\text{alive}) + qW(\text{dead}) \\ &= A(1 + R_b) - qF + (1 - q)H + qF \\ &= A(1 + R_b) + (1 - q)H\end{aligned}$$

- Expected wealth does not depend on F .

Variance of wealth

- Variance of wealth is

$$\begin{aligned}\text{Var}(W) &= (1 - q)(W(\text{alive}) - \mathbb{E}[W])^2 + q(W(\text{dead}) - \mathbb{E}[W])^2 \\ &= (1 - q) \underbrace{(q(H - F))^2}_{\text{alive}} + q \underbrace{((1 - q)(H - F))^2}_{\text{dead}} \\ &= q(1 - q)(H - F)^2\end{aligned}$$

- Variance minimized when $F = H$.

Optimal life insurance

- Expected wealth does not depend on face amount.
- Thus, optimal face amount depends only on the variance of wealth.
- Variance minimized when face amount equal to human capital.
- Intuition:
 - H is the present value of income that is lost if the policyholder dies.
 - If $F = H$, life insurance replaces exactly the income that is lost.

Example 1 (continued)

- Recall the table for expected annual income of \$55,000.

Age	Years to retirement	Human capital (\$)
35	30	1,258,437
45	20	927,800
55	10	510,522
65	0	0

- A 35 year old with annual income of \$55,000 should buy approximately \$1.3 million of life insurance.
- With annual income of \$110,000, \$2.6 million of life insurance (and so on).

Term life insurance

- Annual renewable life insurance insures you for only the next year.
- Instead, you could buy 10-, 20-, or 30-year term life insurance.
- With 20-year term life insurance,
 - Pay annual (or monthly) premium if alive.
 - If you die at any time during the 20-year coverage period, the insurance company pays your beneficiaries the face amount.
- Premium depends on gender and health: preferred (plus), standard (plus), smoker.

Quotes on 20-year term life insurance

- 30-year old male with standard rating (non-smoker).
- Annual premium per \$1 million face amount.

Insurance company	Premium (\$)
Centrian Life Insurance	820
Savings Bank Life Insurance Co of MA	820
Ohio National Life Assurance Corporation	835
⋮	
Midland National Life Insurance Company	1,350
Western-Southern Life Assurance Company	1,560
Provident Life and Accident Assurance	1,580

Source: COMPULIFE Software (12/2012)

When should you buy life insurance?

- When you have dependents (non-working spouse or children) that rely on your income.
- Good idea to buy life insurance when you get married and plan for kids.
- Don't wait too long because life insurance becomes expensive once you get sick unexpectedly (e.g., cancer).
- If you're healthy, individual life insurance is usually cheaper than group life insurance that you can buy through your employer.

Fidelity Freedom Funds

- An example of target date funds.
- Select a fund based on your target retirement date.
 - Fidelity Freedom 2030 Fund.
 - Fidelity Freedom 2035 Fund.
 - Fidelity Freedom 2040 Fund.
 - Fidelity Freedom 2045 Fund.
 - Fidelity Freedom 2050 Fund.
 - Fidelity Freedom 2055 Fund.
 - Fidelity Freedom 2060 Fund.
- Expense ratio of 0.75%.
- No need to worry about optimal portfolio choice or rebalancing. Fidelity does all the work for you!

How do Fidelity Freedom Funds work?



Review of optimal portfolio choice

- Investor has financial wealth A to optimally allocate between stocks S and a riskless asset $A - S$.
- Normalize financial wealth to $A = 1$.
- Portfolio return is

$$\begin{aligned} 1 + R_p &= (A - S)(1 + R_b) + S(1 + R_s) \\ &= \left(\underbrace{\left(1 - \frac{S}{A}\right)}_{1-w} (1 + R_b) + \underbrace{\frac{S}{A}}_w (1 + R_s) \right) \underbrace{A}_{=1} \end{aligned}$$

where

- Riskless rate R_b is constant.
- Stock return R_s has mean μ and variance σ^2 .

- Investor maximizes mean-variance expected utility.

$$\max_w \mathbb{E}[R_p] - \frac{\gamma}{2} \text{Var}(R_p)$$

- Optimal portfolio is

$$w = \frac{S}{A} = \frac{\mu - R_b}{\gamma \sigma^2}$$

Adding human capital

- Investor has financial wealth A to optimally allocate between stocks S and a riskless asset $A - S$.
- In addition, investor has human capital H that earns riskless interest.
- Normalize total wealth to $A + H = 1$.
- Portfolio return is

$$\begin{aligned} 1 + R_p &= (A - S)(1 + R_b) + S(1 + R_s) + H(1 + R_b) \\ &= (A + H - S)(1 + R_b) + S(1 + R_s) \\ &= \left(\underbrace{\left(1 - \frac{S}{A + H}\right)}_{1-w} (1 + R_b) + \underbrace{\frac{S}{A + H}}_w (1 + R_s) \right) \underbrace{(A + H)}_{=1} \end{aligned}$$

- Investor maximizes mean-variance expected utility.

$$\max_w \mathbb{E}[R_p] - \frac{\gamma}{2} \text{Var}(R_p)$$

- Optimal portfolio is

$$w = \frac{S}{A + H} = \frac{S}{A(1 + H/A)} = \frac{\mu - R_b}{\gamma \sigma^2}$$

- As a share of financial wealth,

$$\frac{S}{A} = \left(1 + \frac{H}{A}\right) \frac{\mu - R_b}{\gamma \sigma^2}$$

- Human capital is like a non-tradable bond.
- If you have more human capital, you should invest a higher share of financial wealth in stocks.

Example 2 from “Risk and Return”

- Optimal portfolio for an investor with risk aversion $\gamma = 10$.

$$\begin{aligned}\frac{S}{A} &= \frac{\mu - R_b}{\gamma \sigma^2} \\ &= \frac{0.94\% - 0.08\%}{10(5.33\%)^2} = 0.30\end{aligned}$$

- However, this assumes no human capital.
- Suppose that financial wealth is \$340,000 and human capital is \$510,522.

$$\begin{aligned}\frac{S}{A} &= \left(1 + \frac{H}{A}\right) \frac{\mu - R_b}{\gamma \sigma^2} \\ &= \left(1 + \frac{510,522}{340,000}\right) 0.30 = 0.75\end{aligned}$$

Example 1 (continued): Portfolio choice

- Expected annual income of \$55,000.
- Retirement at age 65.
- Financial wealth of \$340,000.

Age	Years to retirement	Human capital (\$)	Weight
35	30	1,258,437	1.41
45	20	927,800	1.12
55	10	510,522	0.75
65	0	0	0.30

Is it a good idea to buy target date funds?

- Fidelity Freedom 2060 Fund has an expense ratio of 0.68%.
- Complex fund of funds that invests in 30 different Fidelity Series funds, including Growth Company Fund (active) and Large Cap Value Index Fund (passive).
- Does one size fit all?
 - Ratio of human capital to financial wealth varies across investors with the same target retirement date.
 - Large differences in pay by education (advanced degrees), occupation, and geography.
 - You can do better by computing the optimal portfolio yourself and avoiding the fees!

Summary

- Life insurance is important for insuring human capital if you have dependents.
- Common theme with the lecture on “Diversification”: No one should bear idiosyncratic risk.
- Shift your portfolio from stocks to bonds as you age.
- Beware of target date funds. You can save fees and hold a more customized portfolio.